

ECO 362 Financial Investments

Precept 4: Diversification

September 29 – October 3, 2025

- 1 **Statistics Refresher**
- 2 Diversification
- 3 Efficient Frontier
- 4 Tangency Portfolio
- 5 Appendix: Compute Matrices in Excel

Mean and Variance Estimation

- Consider that you observe the returns of I assets over T periods. Let r_{it} be the return of asset i in period t
- Write as a vector

$$\mathbf{r}_t = \begin{bmatrix} r_{1t} \\ \vdots \\ r_{It} \end{bmatrix}$$

- Estimated mean is

$$\hat{\mu} \equiv \frac{1}{T} \sum_{t=1}^T \mathbf{r}_t \quad I \times 1 \text{ vector}$$

- Estimated covariance matrix is

$$\hat{\Sigma} \equiv \frac{1}{T} \sum_{t=1}^T (\mathbf{r}_t - \hat{\mu})(\mathbf{r}_t - \hat{\mu})' \quad I \times I \text{ matrix}$$

- The element (i, j) is the estimated covariance of asset r_i and r_j
- Assume: the mean and covariance matrix remain the same in future

- Suppose that X, Y are random variables and a, b are constants
- Mean scales linearly:

$$\mathbb{E}[aX] = a\mathbb{E}[X]$$

- Variance scales quadratically:

$$\text{Var}(aX) = a^2 \text{Var}(X)$$

- Variance of sum:

$$\text{Var}(aX + bY) = a^2 \text{Var}(X) + b^2 \text{Var}(Y) + 2ab \text{Cov}(X, Y)$$

- Covariance with self is variance:

$$\text{Cov}(X, X) = \text{Var}(X)$$

- Data Analysis ToolPak
- Matrix functions in Excel
 - To execute matrix functions, highlight target cells that are the correct dimension and press Ctrl+Shift+Enter.
- More detail in appendix [▶ Appendix](#)

- 1 Statistics Refresher
- 2 Diversification**
- 3 Efficient Frontier
- 4 Tangency Portfolio
- 5 Appendix: Compute Matrices in Excel

- Assume that the return on stock i is

$$r_i = r_m + \epsilon_i$$

- r_m is the market return, which represents the *systematic* risk
- ϵ_i is the *idiosyncratic* risk of stock i
- ϵ_i has mean 0 and variance σ_ϵ^2 and ϵ_i is uncorrelated with r_m and ϵ_j for $j \neq i$, that is,

$$\text{Cov}(\epsilon_i, \epsilon_j) = \text{Cov}(\epsilon_i, r_m) = 0$$

- Consider an equal-weighted portfolio

$$r_p = \frac{1}{I} \sum_{i=1}^I r_i = r_m + \frac{1}{I} \sum_{i=1}^I \epsilon_i$$

- The mean is $\mathbb{E}(r_p) = \mathbb{E}(r_m)$
- The variance is

$$\begin{aligned}\text{Var}(r_p) &= \text{Var}(r_m) + \frac{2}{I} \sum_{i=1}^I \text{Cov}(r_m, \epsilon_i) + \frac{1}{I^2} \sum_{i=1}^I \sum_{j=1}^I \text{Cov}(\epsilon_i, \epsilon_j) \\ &= \text{Var}(r_m) + \frac{1}{I^2} \sum_{i=1}^I \text{Cov}(\epsilon_i, \epsilon_i) \\ &= \text{Var}(r_m) + \frac{1}{I} \sigma_\epsilon^2 \rightarrow \text{Var}(r_m) \text{ when } I \rightarrow +\infty\end{aligned}$$

- 1 Statistics Refresher
- 2 Diversification
- 3 Efficient Frontier**
- 4 Tangency Portfolio
- 5 Appendix: Compute Matrices in Excel

- Consider portfolios of I risky assets (not including riskless asset)
- What is a **mean-variance efficient** portfolio?
 - has **the highest expected return** among portfolios with **the same variance**
 - has **the smallest variance** among portfolios with **the same expected return**
- What is the **efficient frontier**?
 - A curve that connects all portfolios that are mean-variance efficient.

Efficient Frontier (2 assets)

- Two assets with returns r_A and r_B and variances σ_A^2 and σ_B^2
- Form portfolio with weight w in A and $1 - w$ in B . Then return of portfolio is

$$r_p = wr_A + (1 - w)r_B$$

- What is variance of portfolio? Use formula for variance of sum of random variables:

$$\text{Var}(r_p) = w^2\text{Var}(r_A) + (1 - w)^2\text{Var}(r_B) + 2w(1 - w)\text{Cov}(r_A, r_B)$$

- Write $\text{Cov}(r_A, r_B) = \sigma_A\sigma_B\rho_{AB}$ where ρ_{AB} is correlation between A and B .
Therefore

$$\sigma_p^2 = w^2\sigma_A^2 + (1 - w)^2\sigma_B^2 + 2w(1 - w)\sigma_A\sigma_B\rho_{AB}$$

- Maximum diversification? Minimum diversification?

- Summary: given w , we can compute expected return and variance of portfolio:

$$\mu_p = w\mu_A + (1 - w)\mu_B$$

$$\sigma_p^2 = w^2\sigma_A^2 + (1 - w)^2\sigma_B^2 + 2w(1 - w)\sigma_A\sigma_B\rho_{AB}$$

- Vary w to trace out a curve on the plot of expected returns against volatility
- This curve is called the *efficient frontier* (two asset case)

Lecture 4 Diversification: Example 2

Compute Efficient Frontier for USA and Denmark

- 1 Statistics Refresher
- 2 Diversification
- 3 Efficient Frontier
- 4 Tangency Portfolio**
- 5 Appendix: Compute Matrices in Excel

Optimal portfolio choice problem with I assets

- Now we have I risky assets as well as a riskless asset.
- I stocks with expected return (mean) μ ($I \times 1$) and covariance matrix Σ ($I \times I$)
- Let w be the vector of weights ($I \times 1$)
- Expected return and variance of the portfolio:

$$\mu_p = w' \mu \quad (1)$$

$$\sigma_p^2 = w' \Sigma w \quad (2)$$

- We can construct the efficient frontier with I assets.
- What is a **tangency** portfolio?
 - A portfolio on the efficient frontier that has the highest Sharpe ratio.

$$\frac{\mathbb{E}[r_p] - r_b}{\sigma(r_p)}$$

- Why named "tangency"?
 - Just find the tangent line from the riskless asset with the efficient frontier
- The tangent point on the efficient frontier (consisting of I risky assets)

$$\hat{w} = \frac{1}{\hat{\gamma}} \Sigma^{-1} (\mu - r_b \mathbf{1}) \quad (3)$$

$$\text{where } \hat{\gamma} = \mathbf{1}' \Sigma^{-1} (\mu - r_b \mathbf{1}) \quad (4)$$

- What is a **capital allocation line**?
 - the line that connects the tangency portfolio and the riskless rate.

$$\mathbb{E}[r_p] = r_b + \frac{\mathbb{E}[\hat{r}_p] - r_b}{\sigma(\hat{r}_p)} \sigma(r_p)$$

where $\hat{r}_p$ denotes the tangency portfolio

- Why important?
 - Regardless of the risk aversion, the optimal portfolio choice lies on the capital allocation line

Optimal Portfolio Problem

- Maximizes mean-variance expected utility with γ

$$\mathbb{E}[U] = \mathbb{E}[r_p] - \frac{\gamma}{2} \text{Var}(r_p) = r_b + \mathbf{w}'(\boldsymbol{\mu} - r_b \mathbf{1}) - \frac{\gamma}{2} \mathbf{w}' \boldsymbol{\Sigma} \mathbf{w} \quad (5)$$

- First order condition

$$\boldsymbol{\mu} - r_b \mathbf{1} - \gamma \boldsymbol{\Sigma} \mathbf{w} = 0$$

- Optimal weights in risky assets:

$$\mathbf{w} = \frac{1}{\gamma} \boldsymbol{\Sigma}^{-1} (\boldsymbol{\mu} - r_b \mathbf{1}) \quad (6)$$

- Optimal weight in riskless asset:

$$1 - \mathbf{w}' \mathbf{1} = 1 - \sum_{i=1}^I w_i$$

- The tangency portfolio has weight vector $\hat{\mathbf{w}}$ and risk aversion $\hat{\gamma}$

Optimal Portfolio Problem

- For the tangency portfolio weights $\hat{w}$,
 - Equation 6 should be applied.

$$\hat{w} = \frac{1}{\hat{\gamma}} \Sigma^{-1} (\mu - r_b \mathbf{1}) \quad (7)$$

- There is one more constraint for $\hat{w}$.

$$\mathbf{1} - \hat{w}' \mathbf{1} = 0 \quad (8)$$

- Solve for $\hat{\gamma}$

$$\mathbf{1} = \hat{w}' \mathbf{1} = \mathbf{1}' \hat{w} \quad (9)$$

$$= \frac{1}{\hat{\gamma}} \mathbf{1}' \Sigma^{-1} (\mu - r_b \mathbf{1}) \quad (10)$$

$$\hat{\gamma} = \mathbf{1}' \Sigma^{-1} (\mu - r_b \mathbf{1}) \quad (11)$$

Computing Tangency Portfolio in Excel: Steps

- As shown in Lecture 4 Slides

- Steps:

- 1 Compute:

$$\tilde{\mathbf{w}} = \Sigma^{-1}(\mu - R_b \mathbf{1})$$

- 2 Compute sum of weights:

$$\hat{\gamma} = \mathbf{1}' \tilde{\mathbf{w}} = \sum_{i=1}^I w_i$$

- 3 Divide:

$$\hat{\mathbf{w}} = \frac{1}{\hat{\gamma}} \tilde{\mathbf{w}}$$

Lecture 4 Diversification: Example 3
Compute tangency portfolio for all countries

- Every optimal portfolio is a combination of the riskless asset and the tangency portfolio

$$\begin{aligned} w &= \frac{1}{\gamma} \Sigma^{-1} (\mu - R_b \mathbf{1}) \\ &= \frac{\hat{\gamma}}{\gamma} \underbrace{\frac{1}{\hat{\gamma}} \Sigma^{-1} (\mu - R_b \mathbf{1})}_{\hat{w}} \end{aligned}$$

- Implication: All investors should hold the same portfolio of risky assets, and depending on risk aversion, should allocate different shares to the riskless asset

- 1 Statistics Refresher
- 2 Diversification
- 3 Efficient Frontier
- 4 Tangency Portfolio
- 5 Appendix: Compute Matrices in Excel**

Install Data Analysis ToolPak add-in

- Required to estimate covariance/ correlation from data in excel.
- Windows:
 - See <https://support.microsoft.com/en-us/office/use-the-analysis-toolpak-to-perform-complex-data-analysis-6c67ccf0-f4a9-487c-8dec-bdb5a2cefab6>
- Mac:
 - Click Tools → Add-ins
 - Select Analysis Tool Pak, Click OK.

- Matrix function takes a range of cells for output.
- Type functions in following steps:
 - Select(keep area highlighted) output area based on output matrix size.
 - type "=", continue to take formula, use mouse to select input matrix whenever necessary.
 - Once formula finished. Click "Ctrl+Shift+Enter" on windows or "Command+Shift+Enter" in Mac
- Also see websites mentioned in lectures notes

Matrix inverse, transpose, multiply

- “inverse”: MINVERSE(Input)
- “transpose”: TRANSPOSE(Input)
- “multiply”: MMULT(Input 1, Input 2)